COLLEGE EVENT MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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(241501072)

in partial fulfilment of the End Semester Examination for the course

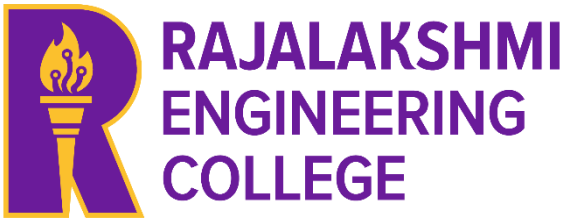
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled **"COLLEGE EVENT MANAGEMENT SYSTEM"** is the bonafide work of **JANANEJAIPRATHA S (241501072)**, who carried out the individual project work on the Zoho Creator low-code platform for the course **CB23F35 – Low-Code Application Development using Zoho Creator**, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, JANANEJAIPRATHA S (241501072), hereby declare that the project report titled "COLLEGE EVENT MANAGEMENT SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report—including its data model, forms, reports, pages, Deluge scripts and workflows—was designed and developed by me. I further declare that any template, AI-assisted feature, code snippet or external resource used in this work has been acknowledged in the report.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Thandalam,Tamil nadu
Date:

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JANANEJAIPRATHA S

ABSTRACT

Colleges conduct many events for students, such as technical symposiums, cultural fests, hackathons, workshops, sports meets and guest lectures. When these registrations are collected on paper, through phone calls or in separate spreadsheets, they are easily lost, there is no common view of their status, and administrators cannot tell which registrations are overdue or how satisfied students are with the events.

This project develops a College Event Management System using Zoho Creator. The application contains six related forms: Student Management, Coordinator Management, Event Management, Event Registration, Payment Management and Feedback Management. Lookup fields connect the records, while field rules and Deluge scripts validate input and calculate the confirmation deadline of each registration from the confirm days of its event. Workflows send acknowledgement, assignment and attendance emails, a scheduled workflow escalates overdue registrations every morning, and an approval process controls inter-college event registrations. A Zoho OAuth connection is used to raise speaker alerts in Zoho Cliq and to read registration data through the Creator REST API. Zia AI fields analyse the sentiment of feedback comments.

Role-based access is configured for Administrator, Event Manager, Event Coordinator and Student (portal) users, and students see only their own records. A dashboard with summary tiles and four charts, together with six reports, gives a consolidated view of registrations, coordinator workload and fee collection. The application is available through the browser, the customer portal and the Zoho Creator mobile app. The result is a single low-code application that replaces scattered registers with traceable, role-controlled and automated campus event handling.

Keywords: Campus Event Management; Zoho Creator; Low-Code Development; Deluge Scripting; Workflow Automation; Role-Based Access Control; Zia AI

PROJECT AT A GLANCE

Attribute	Details
Project Title	College Event Management System
Application Domain	Education / Campus Administration and Student Events
Student Name & Register No.	Jananejaipratha S – 241501072
Zoho Creator Edition	Zoho Creator (trial / evaluation edition)
Application Name (Link Name)	College_Event_Management_System
Published / Live URL	Provided during the demonstration (trial account expired)
Demo Credentials (per role)	Not included in the report; shared securely during the demonstration
Number of Forms / Subforms	6 / 0
Number of Reports (by type)	6 – List (4), Kanban (1), Summary (1)
Number of Pages / Dashboards	1 dashboard page (4 summary tiles and 4 charts)
Number of Workflows	8 – 5 form workflows, 1 schedule, 1 approval process, 1 Blueprint
Deluge Functions / Custom Actions	6 scripts (DS-01 to DS-06)
Integrations	Zoho OAuth connections: Zoho Cliq (invokeurl) and Zoho Creator REST API v2 (tested in Postman)
Roles & Profiles Defined	Administrator, Event Manager, Event Coordinator, Student (portal)
Environments Used	Development, Stage, Production
Date of Production Deployment	October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
Deluge	Data Enriched Language for the Universal Grid Environment
ER	Entity–Relationship
FR / NFR	Functional / Non-Functional Requirement
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SLA	Service Level Agreement
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho's AI assistant (Zoho Intelligence Assistant)
Zoho Creator	Low-code application development platform

CHAPTER 1
INTRODUCTION

1.1 Background

The College Event Management System is a business application for a college campus that must handle many kinds of student event registrations in an organised way. Students want to take part in technical symposiums, cultural fests, hackathons, workshops, sports meets and guest lectures. Each event is owned by a coordinator, and each has its own date, venue, fee and confirmation time. The application uses Zoho Creator forms for data entry, lookup fields to connect records, reports and a dashboard for monitoring, workflows for notifications and an AI-assisted feedback form. It can be accessed from a browser, the student portal and the mobile app, according to the permissions of each user.

1.2 Problem Statement

A campus event desk must receive each registration, identify its event and participation type, assign it to the right coordinator, track its progress against a deadline, collect any event fee and finally capture the student's feedback. When these steps are maintained on paper slips, registers or disconnected spreadsheets, the following problems arise.

- Registrations may be lost, duplicated or forgotten because there is no single record of every registration.
- Students cannot see the status of their registrations and must visit or call the office repeatedly.
- Administrators cannot easily identify overdue registrations, overloaded coordinator or the fees still pending.
- Feedback is not captured in a structured way, so event quality cannot be measured.

1.3 Objectives

- To create related forms for students, event coordinators, events, event registrations, payments and feedback.
- To automate input validation, due-date calculation and selected notifications using Deluge scripts and workflows.

- To provide reports and a dashboard for reviewing registration status, coordinator workload, fee collection and student feedback.
- To integrate the application with Zoho Cliq and the Creator REST API through secure OAuth connections, and to apply Zia AI to analyse feedback comments.
- To apply role-based access for Administrator, Event Manager, Event Coordinator and Student users and verify it through testing.
- To deploy the application through Development, Stage and Production environments with security and governance controls.

1.4 Scope of the Project

In scope: student, coordinator, event, event registration, payment and feedback records; lookup relationships; validation; due-date calculation from confirm days; reports; dashboard and charts; email and Zoho Cliq notifications; scheduled overdue escalation; approval of inter-college event registrations; role-based access and a student portal; Zia sentiment analysis on feedback; mobile access; and deployment across environments.

Out of scope: online payment-gateway processing, biometric or ID-card integration, integration with the college ERP or examination system, and production-scale performance testing with very large datasets.

1.5 Justification for a Low-Code Approach

Zoho Creator is suitable for this application because its visual form builder, lookup fields, reports, pages, workflows and Deluge scripting allow related business records and automation to be built quickly without developing a separate database, back-end server and user interface. Built-in hosting, role and profile management, a customer portal, a mobile app and connections to other Zoho services reduce infrastructure effort, so more time can be spent on the event-management process itself.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and literature. Chapter 3 presents the requirement analysis and Chapter 4 the system design. Chapter 5 details the implementation, Chapter 6 testing, Chapter 7 deployment and security, Chapter 8 results and Chapter 9 the conclusion.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a manual process, a student visits the department office or notice board and writes his or her name on a registration slip or fills a paper form. A clerk notes the registration in a register or spreadsheet and passes it on verbally or by phone to the responsible coordinator. The coordinator acts on it, but there is no formal status update, so the student returns repeatedly to ask whether the registration is confirmed. Fees for events such as workshops or hackathons are recorded separately, and feedback is rarely collected.

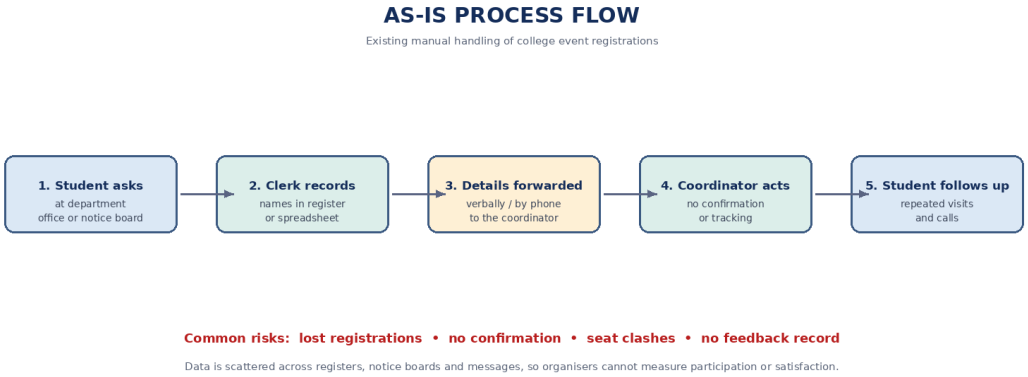


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Records are scattered across registers, slips and spreadsheets, making retrieval and follow-up slow.
- There is no automatic acknowledgement, confirmation deadline or reminder, so urgent registrations can be delayed without anyone noticing.
- Students have no visibility of registration status, and administrators have no summary of workload, overdue items or feedback.
- Fee collection and event quality cannot be reviewed because data is not stored in a structured and connected form.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Requires custom UI, database, backend, testing and hosting setup.	Visual form/report builders and built-in hosting shorten initial development; actual time depends on scope.
Skills required	Programming, database design, deployment and maintenance skills.	Visual configuration plus Deluge and platform knowledge.
Hosting & infrastructure	Application server, database and deployment pipeline must be maintained.	Hosting is provided through the Zoho Creator service.
Mobile application	Usually requires responsive web design or separate mobile development.	Creator provides platform-supported mobile access for the same application.
Security & access control	Must be designed and implemented in application code and infrastructure.	Built-in roles, profiles and portal permissions are available but must be configured and tested.
Maintenance & change	Code and infrastructure changes may require developer effort.	Many form and report changes can be made visually; scripts still need maintenance.
Customisation limits	Broad control over architecture and runtime.	Bounded by Creator features, limits and supported integrations.
Cost	Development, hosting, monitoring and maintenance costs vary.	Subscription/edition and usage costs depend on the selected plan.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S. No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation, Zoho Creator Help	Forms and fields documentation	Creator forms provide configurable field types, validation and data capture.	Supports the six form modules and field rules.

S. No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
2	Zoho Corporation, Zoho Creator Help	Lookup fields and relationships	Lookup fields connect records across forms.	Supports student–registration, category–registration and registration–payment relationships.
3	Zoho Corporation, Deluge Help	Deluge scripting documentation	Deluge provides control flow, record access, sendmail and invokeurl for integrations.	Supports validation, due-date calculation, notifications and API calls.
4	Zoho Corporation, Zoho Creator Help	Workflows, approvals and Blueprint	Workflows run actions on record events or schedules; approvals and Blueprint control process stages.	Supports acknowledgement emails, escalation and the registration lifecycle.
5	Zoho Corporation, Zoho Creator Help	Connections and REST API v2	OAuth connections authorise calls to Zoho services and external APIs without storing passwords in scripts.	Supports the Zoho Cliq alert and the Postman API test.
6	Sahay et al., 2020	Supporting the understanding and comparison of low-code development platforms	Compares features of low-code platforms and highlights visual modelling, integration and deployment support.	Justifies the choice and evaluation of a low-code platform.
7	OWASP Foundation	Application Security Verification Standard (ASVS)	Security should be assessed through access control, input validation and secure handling of data.	Informs permission checks and validation testing.

2.5 Emerging Trends in Low-Code Development

Low-code platforms increasingly include AI-assisted application generation, visual workflow builders, ready-made connectors and AI services such as sentiment analysis and prediction. In this project, an AI assistant helped to build part of the form structure, and Zia sentiment and keyword fields are used to analyse the text of student feedback. AI-generated configuration must always be reviewed against the real requirements, which is why every Zia-generated field and script in this application was checked and tested before use. Integration with collaboration tools such as Zoho Cliq is another trend that turns a data-entry application into a live event desk.

2.6 Proposed System

The proposed system uses six connected forms as the central data model. Student, coordinator and event information is maintained once and referenced through lookup fields in the Event Registration form. Payments and feedback are linked to the registration they belong to. Validation and Deluge scripts calculate confirmation deadlines and block invalid input; workflows acknowledge registrations, notify coordinator, invite feedback and escalate overdue registrations; and an approval process controls inter-college event registrations. Reports and a dashboard summarise registrations, coordinator workload and fees, and the student portal lets students submit and track their own registrations. The effectiveness of each workflow is demonstrated using test records and delivered emails.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Application owner or campus event administrator	Manage all records, users, roles, workflows, configuration and reports.
Event Manager	Head of the events department	Review and assign registrations, monitor the dashboard and approve inter-college event registrations.
Event Coordinator	Coordinator who handles registrations (faculty coordinators, club secretaries, sports in-charge and department staff)	View assigned registrations, update status and attendance, and record payments for paid events.
Student	Student who needs a campus event (portal user)	Submit registrations, track own registrations, view own payments and give feedback.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Participation Type
FR-01	The system shall allow authorised users to create and maintain student and event coordinator records with mandatory contact fields.	Administrator, Manager	High
FR-02	The system shall maintain events with owner department, confirm days and fee.	Administrator	High
FR-03	The system shall allow students to submit event registrations linked to their own record and an event.	Student, Coordinator	High
FR-04	The system shall calculate the confirmation deadline of a registration from its registration date and the confirm days of its category.	System	High
FR-05	The system shall allow registrations to be assigned to coordinator and tracked through Registered, Assigned, Confirmed, Attended and Closed.	Administrator, Manager, Coordinator	High

FR ID	Requirement	Role(s)	Participation Type
FR-06	The system shall record payments for paid events and report pending payments.	Coordinator, Administrator	Medium
FR-07	The system shall send acknowledgement, assignment and attendance emails when workflow conditions are met.	System workflow	High
FR-08	The system shall escalate registrations that pass their confirmation deadline and alert Zoho Cliq for speaker registrations.	System workflow	Medium
FR-09	The system shall capture ratings and comments and analyse comment sentiment using Zia.	Student, System	Medium
FR-10	The system shall display a dashboard with summary tiles and charts on registrations, coordinator workload and fees.	Administrator, Manager	Medium
FR-11	The system shall restrict access to forms, reports and records according to the user's role and profile.	All roles	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard should load within a few seconds for the expected number of campus records.
NFR-02	Security	Users shall access only forms, reports and records allowed by their assigned role and profile; connections use minimum OAuth scopes.
NFR-03	Usability	Forms should use clear labels, required-field indicators and understandable validation messages on desktop and mobile.
NFR-04	Availability	Availability depends on the Zoho Creator service and the user's internet connection.
NFR-05	Maintainability	Forms, fields, scripts and workflows should use meaningful names and be documented for future changes.
NFR-06	Auditability	Record and configuration changes should be traceable through the audit trail.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (trial / evaluation edition)
Scripting language	Deluge scripting
Browser	Current version of Chrome, Edge or Firefox
API testing tool	Postman (to test the Creator REST API v2 with the OAuth connection)
Other Zoho services	Zoho Cliq (alerts), Zoho OAuth connections, Zia AI fields
Mobile	Zoho Creator mobile app (Android / iOS) and student portal on a mobile browser
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

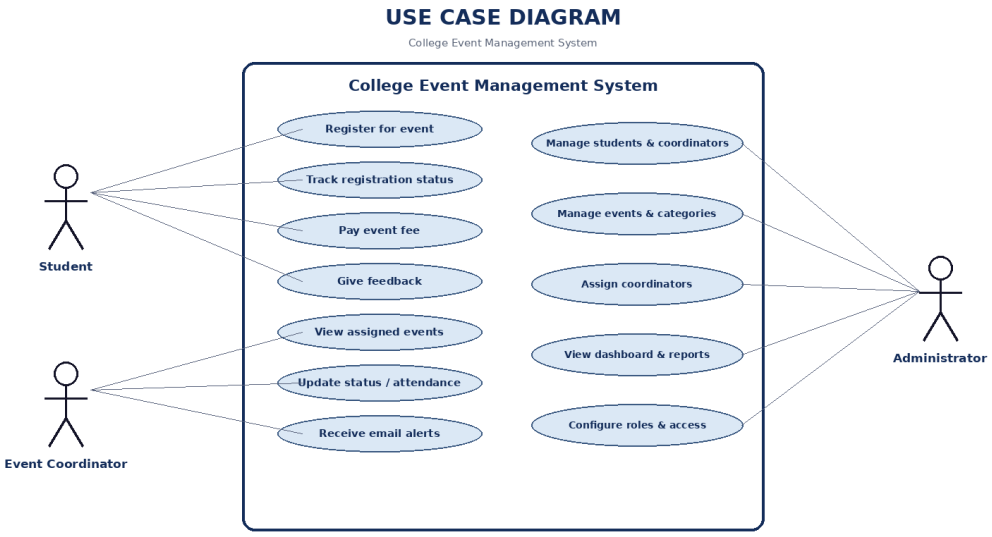


Figure 3.1: Use Case Diagram of the College Event Management System

Table 3.5: Use Case Description – Submit and Complete an Event Registration

Use Case ID	UC-01
Use Case Name	Submit and Complete an Event Registration
Actor(s)	Student, Administrator / Event Manager, Event Coordinator
Pre-condition	The student has a portal account; events and coordinator records exist.
Main Flow	1. Student signs in to the portal and opens Event Registration. 2. Student selects the event, enters subject, description and participation type, and submits. 3. The system sets the confirmation deadline and sends an acknowledgement email. 4. Administrator assigns a coordinator; the coordinator is notified. 5. Coordinator updates the status to In Progress and then Attended. 6. The student receives a attendance email and submits feedback.
Alternate Flow	If a required field or lookup value is missing, the form shows a validation message and the registration is not saved. If the student email is missing, the acknowledgement email is not sent. Speaker registrations also raise a Zoho Cliq alert.
Post-condition	The registration, its status history and the feedback are saved and visible in the reports and dashboard.
Related FR	FR-03, FR-04, FR-05, FR-07, FR-09

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

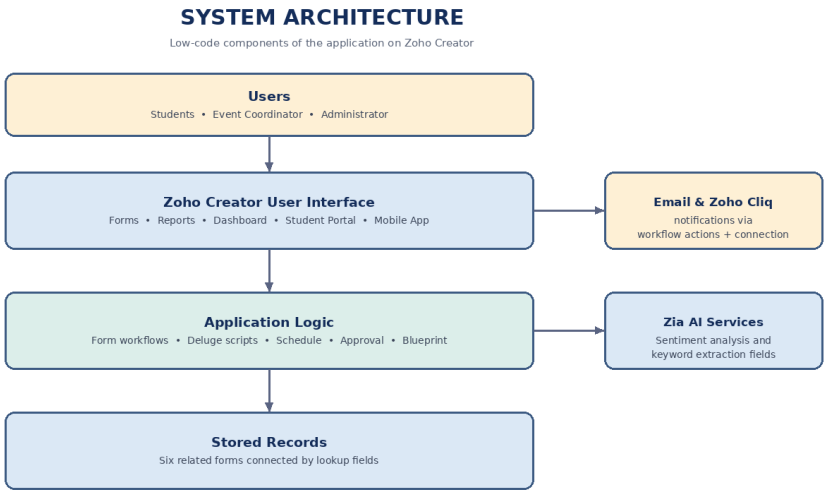


Figure 4.1: System Architecture of the College Event Management System

Users access the Zoho Creator application through its user interface, the student portal or the mobile app. Forms capture and store student, coordinator, category, registration, payment and feedback records, and lookup fields connect the related records. Reports and the dashboard read the stored data for review. Form workflows, Deluge scripts, a schedule, an approval process and a Blueprint implement validation, calculation and notifications. Connections authorise calls to Zoho Cliq and the Creator REST API, and Zia AI fields analyse feedback text. Access to forms, reports and actions is controlled by roles and profiles.

4.2 Data Model

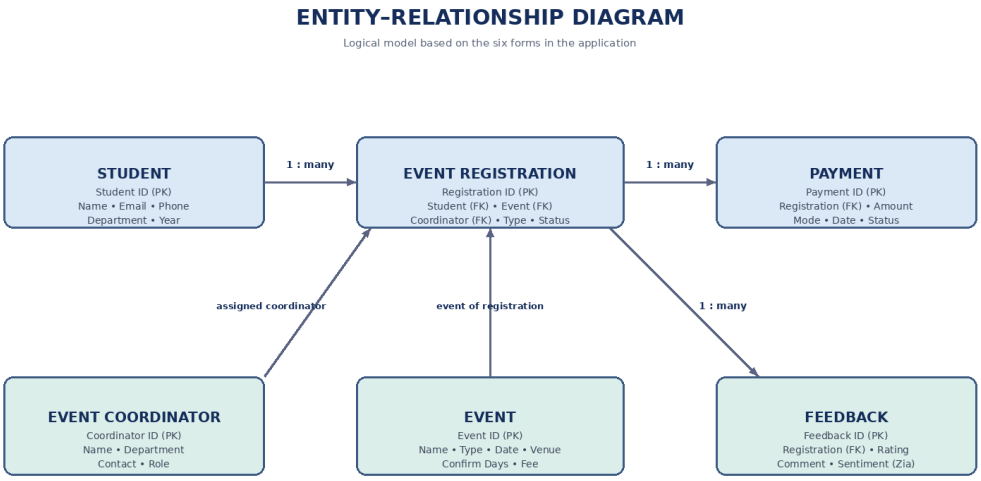


Figure 4.2: Entity–Relationship Diagram of the College Event Management System

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Student Management	Student ID, Student Name, Email	Event Registration	One-to-Many	Student lookup in Event Registration
Event Management	Event ID, Event Name, Confirm Days, Fee	Event Registration	One-to-Many	Event lookup in Event Registration
Coordinator Management	Coordinator ID, Name, Department, Contact	Event Registration	One-to-Many	Assigned Coordinator lookup in Event Registration
Event Registration	Registration ID, Team Name, Participation Type, Status, Confirm By	Payment Management	One-to-Many	Registration lookup in Payment
Event Registration	Registration ID, Team Name, Participation Type, Status, Confirm By	Feedback Management	One-to-Many	Registration lookup in Feedback

No subform is used. Payments and feedback are stored as separate forms because a registration can have several payments and each feedback entry needs its own report and permissions; a subform would have hidden these records from reports.

4.3 Form Design

Table 4.2: Field Specification – Student Campus Event Management Forms

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student ID	Student_ID	Auto Number	Auto	Prefix STU-	Student identity
Student Name	Student_Name	Name	Yes	Required	Student identity
Email	Email	Email	Yes	Valid, no duplicates	Used for notifications
Department	Department	Dropdown	Yes	Required	AIML, CSE, ECE, IT, MECH
Event Name	Event_Name	Single Line	Yes	Required	Event master
Confirm Days	Confirm_Days	Number	Yes	Must be >= 1	Used for confirmation deadline
Event Fee	Event_Fee	Currency	No	Must be >= 0	Paid events
Student	Student	Lookup	Yes	Student Management	Registration owner
Event Management	Event	Lookup	Yes	Event Management	Event being joined
Team Name	Team Name	Single Line	Yes	Required	Team or entry name
Participation Type	Participation Type	Radio	Yes	Default: Medium	Participant / Volunteer / Speaker
Registration Date	Registration_Date	Date	Yes	Default: today	Cannot be blank
Confirm By	Confirm_By	Date	No	Set by DS-01	Registration date + confirm days
Assigned Coordinator	Assigned_Coordinator	Lookup	No	Coordinator Management	Set by administrator
Status	Status	Dropdown	Yes	Default: Registered	Managed by Blueprint
Amount	Amount	Currency	Yes	Must be >= 0	Payment value
Rating	Rating	Radio	Yes	1 to 5	Feedback score
Comment Sentiment	Comment_Sentiment	Zia AI field	No	From Comment	Positive / Neutral / Negative

4.4 Process and Workflow Design

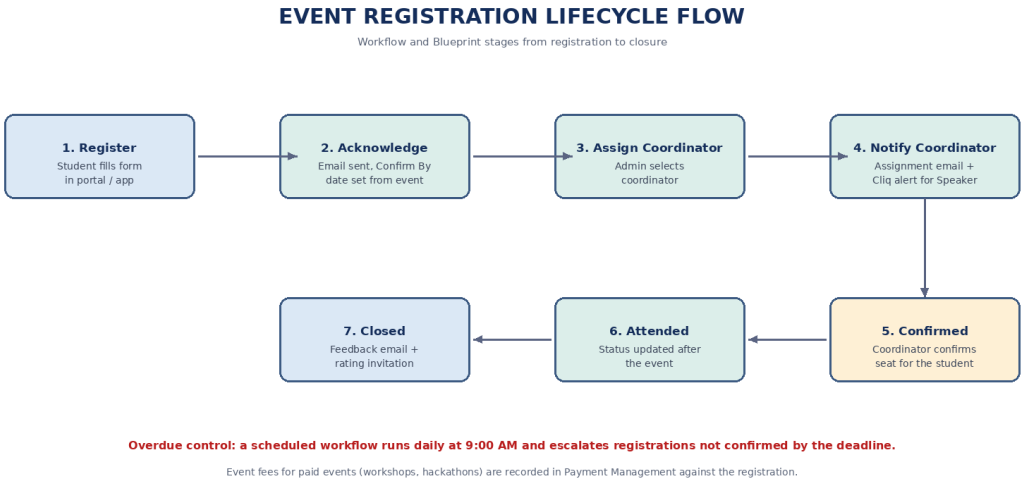


Figure 4.3: Workflow / Process Flow for Event Registration Handling and Notification

4.5 User Interface and Navigation Design

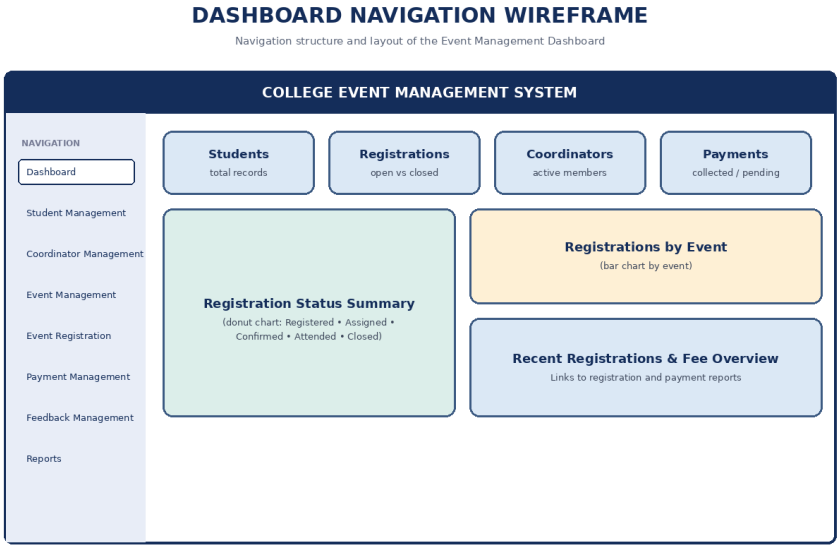


Figure 4.4: Navigation Structure and Wireframe of the Event Management Dashboard

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Administrator	Manager	Coordinator	Student Portal
Form: Student Management	A E V D	A E V	V	—
Form: Coordinator Management	A E V D	A E V	V	—

Component	Administrator	Manager	Coordinator	Student Portal
Form: Event Management	A E V D	E V	V	V
Form: Event Registration	A E V D	A E V	E V (assigned)	A V (own)
Form: Payment Management	A E V D	A E V	A V	V (own)
Form: Feedback Management	A E V D	V	V	A V (own)
Report: All Event Registrations	V	V	V (assigned)	V (own)
Report: Pending Payments	V	V	V	—
Page: Event Management Dashboard	V	V	—	—

A = Add, E = Edit, V = View, D = Delete. “(own)” and “(assigned)” indicate record-level criteria.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “College Event Management System” was created in Zoho Creator with the link name College_Event_Management_System. Its navigation is organised into sections for the six data-entry forms, the reports and the Event Management Dashboard. The home page offers quick links to add records and to open the main reports, and the project uses Creator’s visual form builder, workflows, Deluge scripting, connections and Zia features, as described in the following sections.

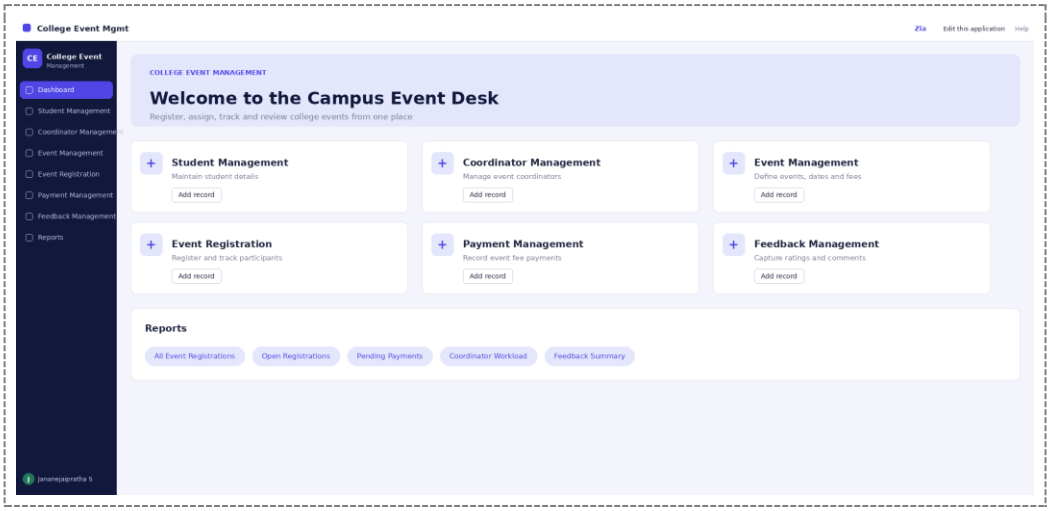


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

The application contains six main forms. Student Management stores identity and contact details; Coordinator Management stores the event coordinators who handle registrations; Event Management defines each event with its coordinator, date, venue, confirm days and fee; Event Registration records every registration with its student, event, participation type, dates, assigned coordinator and status; Payment Management records payments for paid events against a registration; and Feedback Management stores the rating and comment for a registration. Across the forms, more than ten distinct field types are used, including Auto Number, Name, Single Line, Multi Line, Email, Phone, Number, Currency, Date, Dropdown, Radio, Checkbox, Lookup, File Upload, Image and Zia AI fields. Required fields and unique-value checks prevent incomplete or duplicate records.

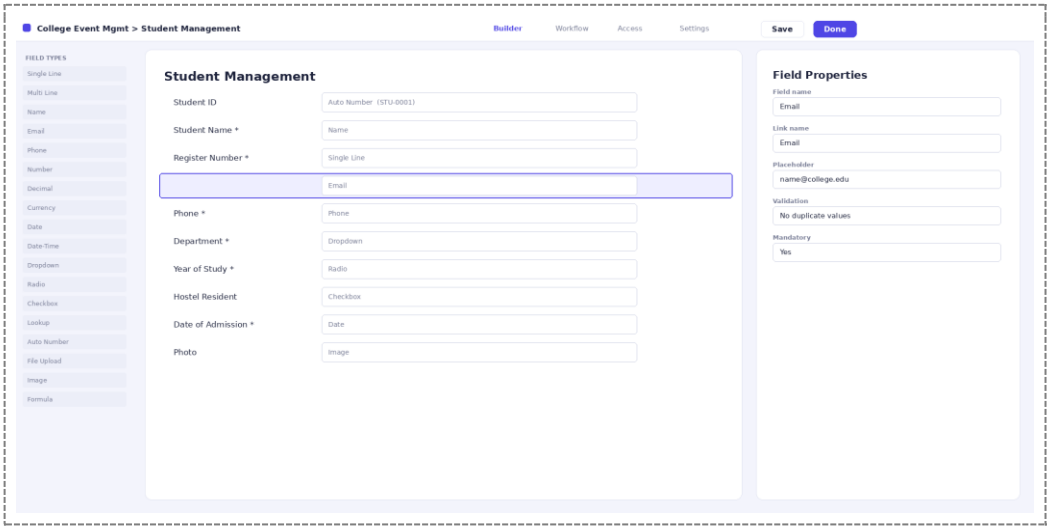


Figure 5.2: Student Management Form in the Form Builder

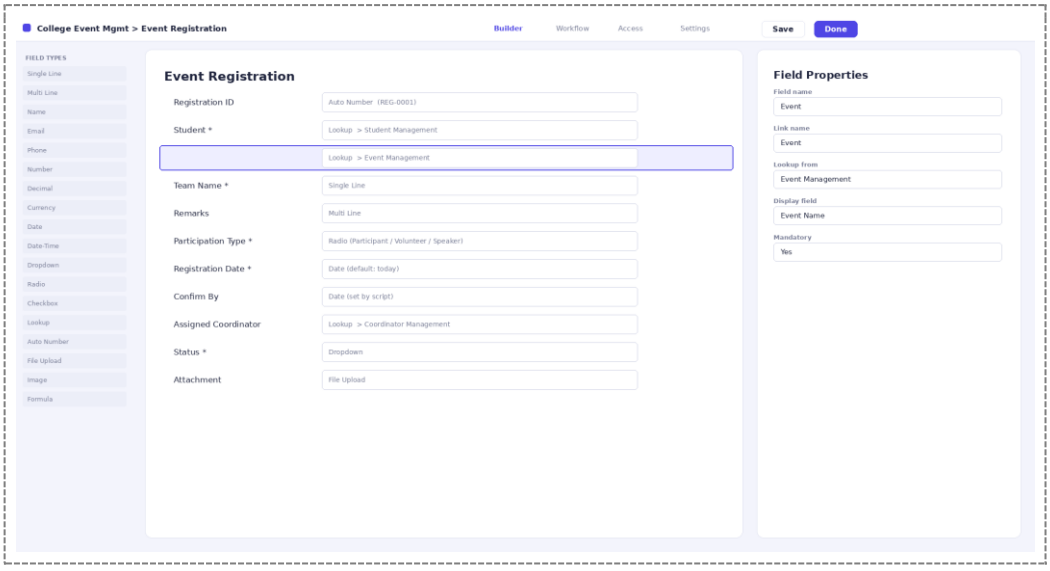


Figure 5.3: Event Registration Form in the Form Builder

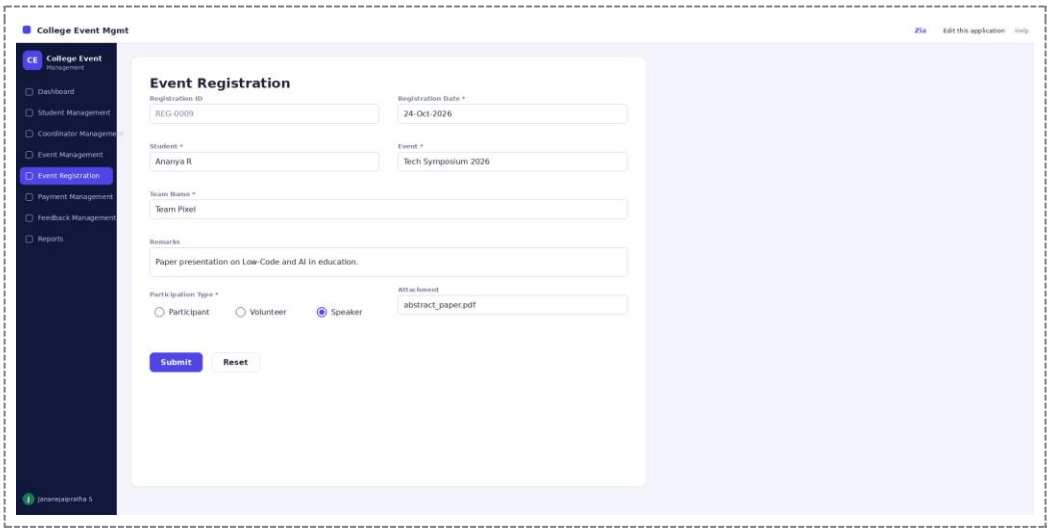


Figure 5.4: Event Registration Form in Live Mode

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Event Registrations	List	Event Registration	Filters: Status, Participation Type, Category, Date; sort by registration date	Open / edit record, export	Administrator, Manager, Coordinator (assigned)
Open Registrations	List	Event Registration	Criteria: Status is not Attended or Closed; sort by confirmation deadline	Assign coordinator	Administrator, Manager
Registration Board	Kanban	Event Registration	Grouped by Status	Drag to change status (Blueprint)	Administrator, Manager, Coordinator
Pending Payments	List	Payment Management	Filter: Status = Pending; sort by date	Open payment record	Administrator, Manager, Coordinator
Coordinator Workload	Summary	Event Registration	Grouped by Assigned Coordinator and Status; count of records	Drill down to records	Administrator, Manager
Feedback Summary	List	Feedback Management	Filter by Rating and Sentiment	Open feedback record	Administrator, Manager

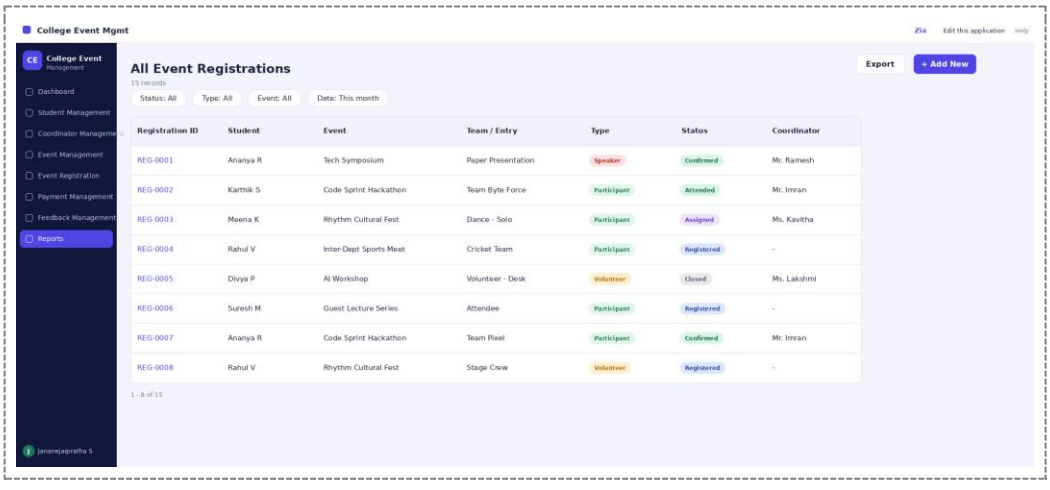


Figure 5.5: All Event Registrations Report

5.4 Pages and Dashboards

The Event Management Dashboard is a page built with the UI builder. It shows summary tiles for total students, event coordinators, events and total registrations,

together with open registrations, attended or closed registrations, fees collected and pending fees. Below the tiles, quick actions open the most common forms. Four charts, created from the Event Registration and Payment forms, show registrations by event (bar), registrations by status (donut), monthly registrations (line) and open registrations by coordinator (horizontal bar). Dashboard values were compared with the underlying reports after records were added or edited.

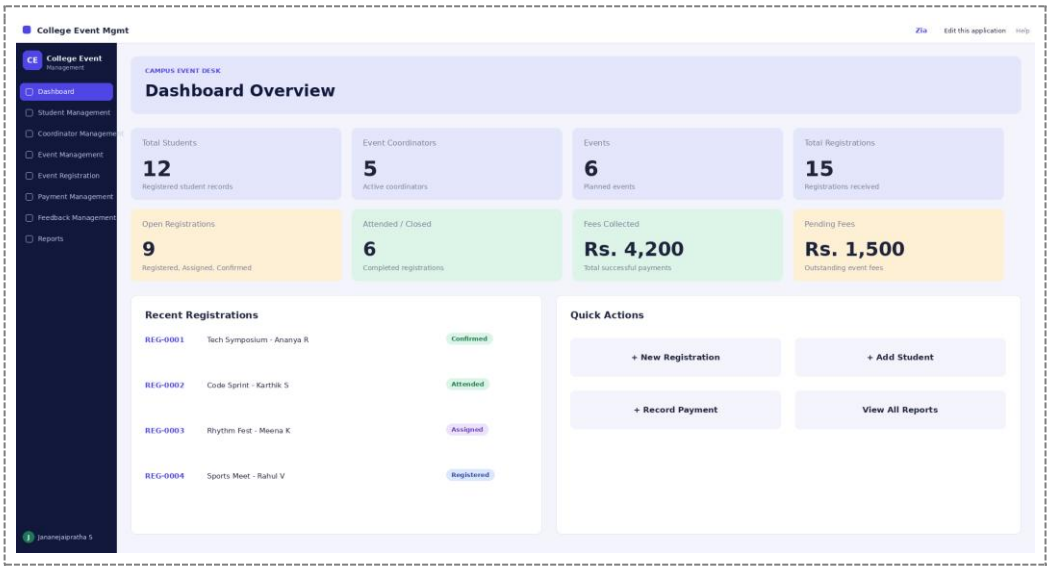


Figure 5.6: Event Management Dashboard – Summary Tiles



Figure 5.7: Event Management Dashboard – Charts

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Event Registration → On Add → On Validate	Validate required values and set the confirmation deadline from the category confirm days.	criteria fetch, if conditions, date functions, alert, cancel submit
DS-02	Event Registration → On Add → On Success	Send an acknowledgement email to the student.	lookup fetch, if conditions, sendmail
DS-03	Event Registration → On Edit → On Success	Notify the assigned coordinator when a registration is assigned.	condition on field change, sendmail
DS-04	Event Registration → On Edit → On Success	Send attendance email with a feedback link when Status becomes Attended.	lookup fetch, if conditions, sendmail
DS-05	Event Registration → On Add → On Success	Post a Zoho Cliq alert for speaker registrations.	Map, invokeurl, connection, info
DS-06	Schedule: daily at 9:00 AM	Email the administrator about registrations past their confirmation deadline.	for each, criteria fetch, date comparison, sendmail

Code 5.1: Validation and Due-Date Calculation Script (DS-01)

```
// On Add > On Validate of Event Registration
cat = Event[ID == input.Event];
if(input.Registration_Date == null)
{
    alert "Registration Date is required";
    cancel submit;
}
if(cat.Confirm_Days == null || cat.Confirm_Days < 1)
{
    alert "confirm days are not defined for this event";
    cancel submit;
}
input.Confirm_By =
input.Registration_Date.addDay(cat.Confirm_Days.toLong());
```

Code 5.2: Acknowledgement Email Script (DS-02)

```
// On Add > On Success of Event Registration
stu = Student_Management[ID == input.Student];
cat = Event[ID == input.Event];
if(stu.Email != "")
{
    sendmail
    [
        from :zoho.adminuserid
        to :stu.Email
        subject :"Registration " + input.Registration_ID + " received"
        message :"Dear " + stu.Student_Name + ", your " +
cat.Event_Name + " registration is registered. Confirmation deadline:
" + input.Confirm_By
    ]
}
```

The validation script checks that the registration date and the confirm days are acceptable before the record is saved. The confirmation deadline is then derived from the category so that students and coordinator see the same deadline. Email scripts fetch the saved student and category records through lookup fields and send a message only when a valid student email exists. The complete source code of all six scripts is given in Appendix C.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Set Confirm By from confirmation time	Form workflow	On Add – On Validate	Registration Date and confirm days available	Calculate Confirm By	DS-01
Registration Acknowledgement Email	Form workflow	On Add – On Success	Student email available	Send acknowledgement email	DS-02
Coordinator Assignment Notification	Form workflow	On Edit – On Success	Assigned Coordinator set and Status is Assigned	Email assigned coordinator	DS-03
Attendance & Feedback Invite	Form workflow	On Edit – On Success	Status becomes Attended	Email student with feedback link	DS-04
Speaker Cliq Alert	Form workflow	On Add – On Success	Participation Type is Speaker	Post alert to Zoho Cliq	DS-05

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Pending Registration Escalation	Schedule	Daily at 9:00 AM	Confirm By earlier than today and not Attended or Closed	Notify administrator	DS-06
Inter-College Event Approval	Approval process	On Add	Category is Inter-College Event	Manager approves or rejects	—
Registration Lifecycle	Blueprint	Status transitions	Registered → Assigned → Confirmed → Attended → Closed	Controls allowed status changes	—



Figure 5.8: Workflow Configuration for Registration Notifications

Code 5.3: Overdue Escalation Script (DS-06)

```
// Schedule: Pending Registration Escalation - daily at 9:00 AM
for each rec in Event_Registration[Confirm_By < zoho.currentdate &&
Status != "Attended" && Status != "Closed"]
{
    sendmail
    [
        from :zoho.adminuserid
        to :zoho.adminuserid
        subject :"Overdue registration " + rec.Registration_ID
        message :"Registration " + rec.Registration_ID + " (" +
rec.Team Name + ") is past its confirmation deadline " +
rec.Confirm_By
    ]
}
```

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Cliq alert	Cliq channel message API (channel: collegeevents)	Deluge invokeurl, POST	Zoho OAuth connection cliq_alert_conn (ZohoCliq.Webhooks .CREATE)	Posts registration ID and subject for speaker registrations
Creator REST API v2	report/All_Event _Registrations	GET (tested in Postman)	Zoho OAuth connection creator_api_conn (ZohoCreator.report. READ)	Returns registration records as JSON for external use
Email notification s	Zoho Creator workflow action	Deluge sendmail	Configured sender account	Acknowledgement, assignment, attendance and overdue emails

Two connections were created under Settings → Connections using the Zoho OAuth service. Each connection stores only the scopes it needs, so no password or token is written in any script. The Cliq connection is referenced by name in the invokeurl call of DS-05, and the Creator API connection authorised the Postman request that returned the JSON response shown in Figure 5.10.

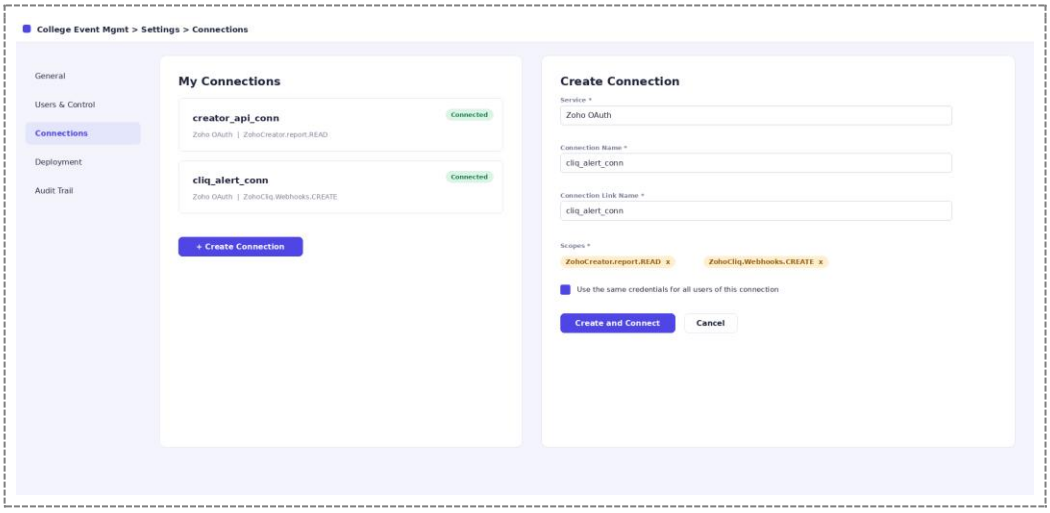


Figure 5.9: Connection Setup in Zoho Creator

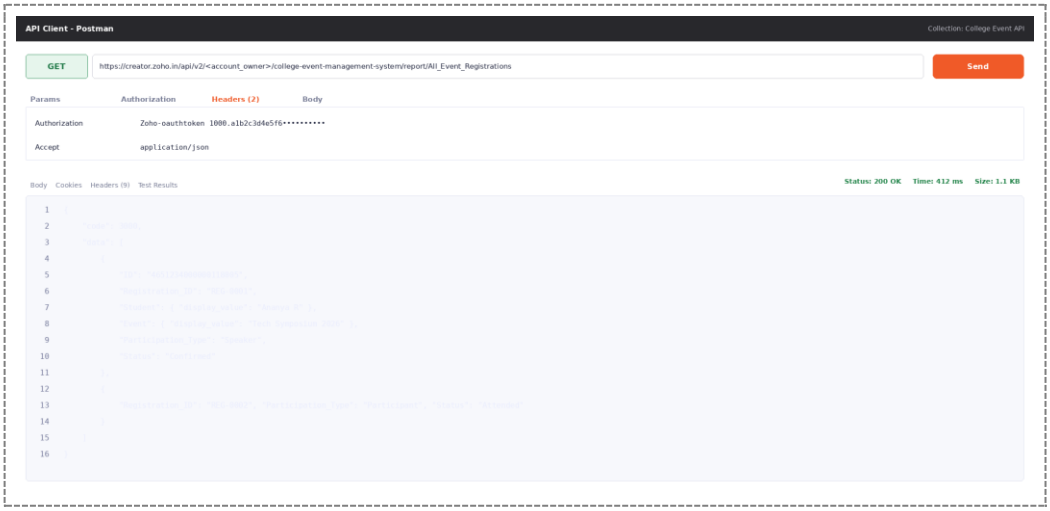


Figure 5.10: Postman Registration and JSON Response from the Creator REST API

Code 5.4: Zoho Cliq Integration Script (DS-05)

```
// DS-05: Speaker alert to Zoho Cliq using a secured connection
if(input.Participation_Type == "Speaker")
{
    payload = Map();
    payload.put("text", "Speaker registration " +
input.Registration_ID + ": " + input.Team Name);
    response = invokeurl
    [
        url
:"https://cliq.zoho.in/api/v2/channelsbyname/collegeevents/message"
        type :POST
        parameters :payload.toString()
        connection :"cliq_alert_conn"
    ];
    info response;
}
```

5.8 User Management and Role-Based Access

Four roles are arranged in a hierarchy: Administrator, Event Manager, Event Coordinator and Student (Portal). Each user is given a profile that decides which forms, reports and pages are visible and which actions are allowed. The Administrator can manage all records and configuration; the Event Manager can add and edit operational records and view the dashboard; Event Coordinator can view and update only the registrations assigned to them and record payments; and students can add and view only their own registrations, payments and feedback. Record-level criteria compare the logged-in user with the Student record, so one student cannot open another student's data. Test accounts for each role were used to verify these permissions.

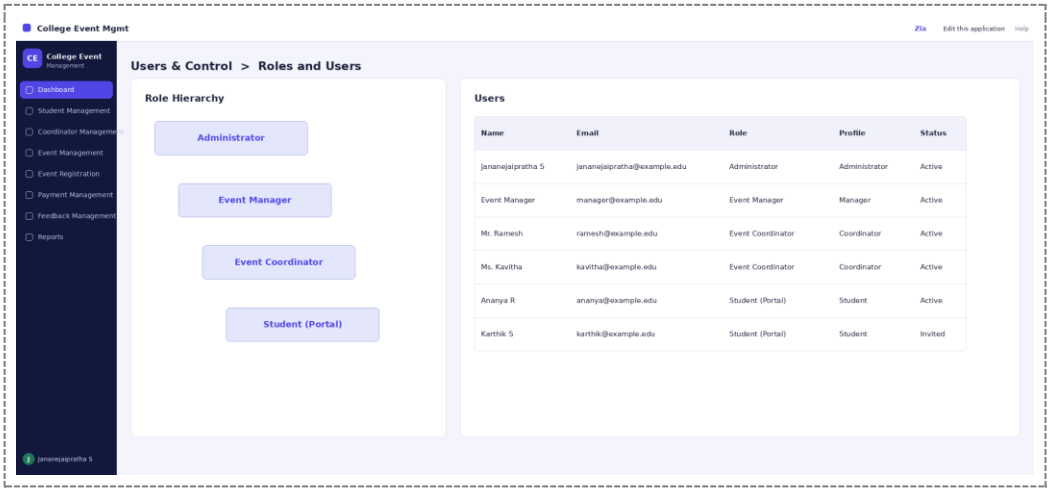


Figure 5.11: Roles and Users Configuration



Figure 5.12: Permission Profile Configuration for Event Coordinator

5.9 AI and Analytics Features

Zia AI features are used in the Feedback Management form. A Sentiment Analysis field reads the Comment text and records whether it is positive, neutral or negative, and a Keyword Extraction field lists the main topics mentioned. The fields were created with the help of the Zia builder assistant and then reviewed and tested before being saved. Administrators can filter the Feedback Summary report by sentiment to find events that need attention.

Analytics are provided by the dashboard charts and the Coordinator Workload summary report. They show registration counts by category and status, the monthly trend of registrations, coordinator workload and fee collection. These are descriptive analytics; no predictive model is claimed.

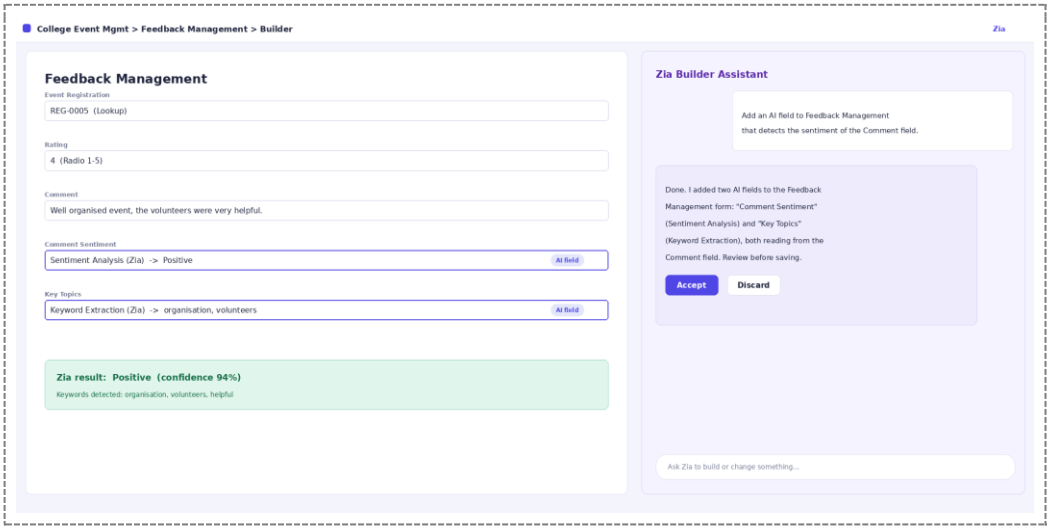


Figure 5.13: Zia AI Fields and Builder Assistant in Feedback Management

5.10 Pro-Code Extension

No pro-code extension such as a Catalyst function, a custom widget or an external microservice was needed. Every requirement was met with forms, workflows, Deluge scripts and connections, which keeps the application easy to maintain by campus coordinatorss who are not professional developers. If a future requirement cannot be met within Creator, such as complex payment verification, a Zoho Catalyst function can be called through the same connection mechanism.

5.11 Mobile and Portal Access

Students use a customer portal with their own sign-in to submit registrations and view their status. Coordinator and administrators can also use the Zoho Creator mobile app. The same forms and reports adapt to the smaller screen, and the permissions defined for each role apply unchanged on every device.

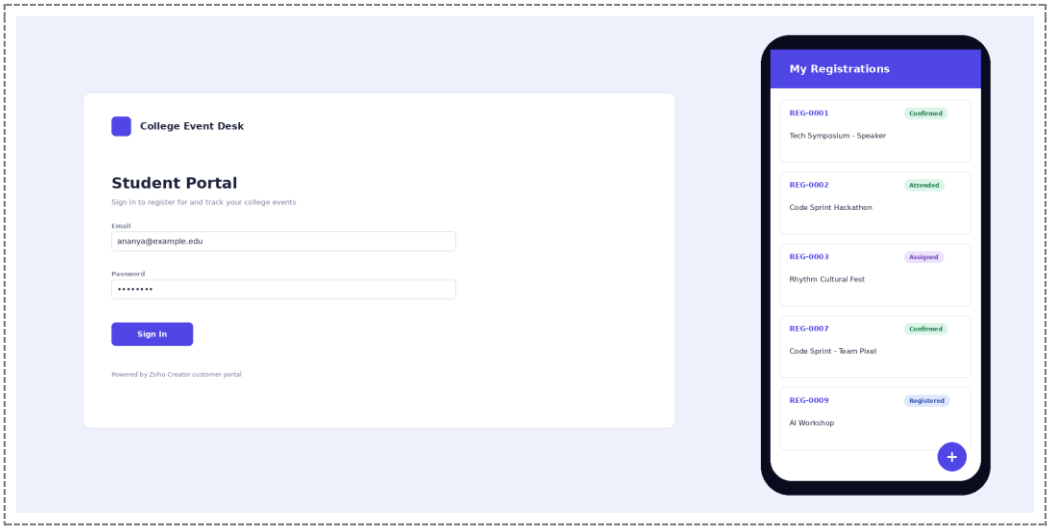


Figure 5.14: Student Portal Sign-In and Mobile View of My Registrations

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was planned at three levels. Form-level tests check required fields, valid data types, date rules and duplicate handling. Integration tests check lookup relationships, workflow triggers, emails, the Cliq alert and the API call. Acceptance tests follow a complete event registration lifecycle with users in each role. Each test case in Table 6.1 is linked to a functional requirement.

6.2 Test Cases

Testing was carried out in a controlled sequence: create valid records, test invalid field values, verify lookup relationships, trigger workflows, inspect reports and the dashboard, and verify role permissions with separate accounts.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Add a student with all mandatory fields	Enter valid name, email, department	Student record saved	Saved as expected	Pass
TC-02	FR-01	Add a student without an email	Leave Email blank	Validation prevents submission	Message shown; not saved	Pass
TC-03	FR-01	Add a student with a duplicate email	Re-use an existing email	Duplicate rejected	Duplicate rejected	Pass
TC-04	FR-03	Submit a registration through the portal	Select category, enter subject and participation type	Registration saved with status New	Saved as New	Pass
TC-05	FR-04	Check due-date calculation	Category confirmation time = 3 days, registration date 24-Oct-2026	Confirmation deadline 27-Oct-2026	Confirmation deadline 27-Oct-2026	Pass
TC-06	FR-04	Category without confirm days	Submit registration for such a category	Submission blocked with alert	Alert shown	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-07	FR-07	Acknowledgement email	Submit a registration for a student with a test email	Email received	Email received	Pass
TC-08	FR-05	Assign coordinator	Set Assigned Coordinator and Status Assigned	Coordinator receives email	Email received	Pass
TC-09	FR-07	Attendance email	Change Status to Attended	Student receives feedback link	Email received	Pass
TC-10	FR-08	Speaker alert	Submit a Speaker registration	Message appears in Cliq channel	Message posted	Pass
TC-11	FR-08	Overdue escalation	Set confirmation deadline in the past and run schedule	Administrator notified	Email received	Pass
TC-12	FR-09	Sentiment analysis	Submit a positive comment	Sentiment = Positive	Positive	Pass
TC-13	FR-11	Student record isolation	Sign in as a student and open the registration report	Only own registrations visible	Only own registrations	Pass
TC-14	FR-11	Coordinator delete permission	Sign in as coordinator and try to delete a registration	Delete option unavailable	Not available	Pass

6.3 Debugging Deluge Scripts

Scripts were debugged by adding info statements that print intermediate values, such as the confirm days and the calculated confirmation deadline, and by reading the output in the script editor. Workflow executions were also checked in the workflow log, and API calls were tested separately in Postman before being used in scripts. Figure 6.1 shows the due-date script with its info output.

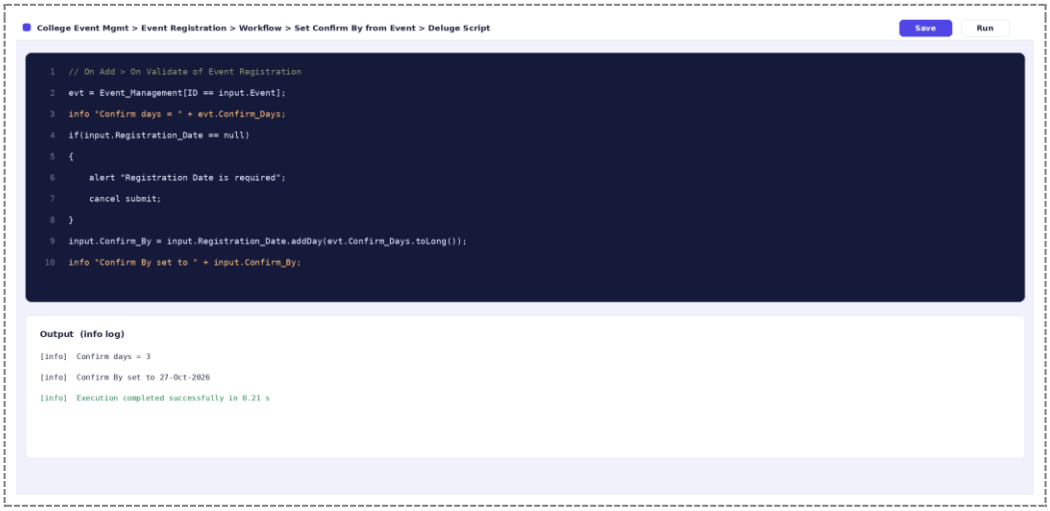


Figure 6.1: Deluge Debug Output for the Due-Date Script

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Confirmation deadline was empty for a new category	Confirm Days was left blank in the category record	Made Confirm Days mandatory and added the check in DS-01	Closed
BUG-02	Coordinator could see all registrations in the report	Profile had no record-level criteria	Added criteria: Assigned Coordinator is the logged-in user	Closed
BUG-03	Attendance email sent again on every edit	Workflow ran for every update	Added condition: Status is Attended	Closed

6.5 User Acceptance Testing

User acceptance testing was done with test accounts for each role. Defects found were added to the defect log, fixed and retested before the application was published. The feedback collected is summarised in Table 6.3.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Test user 1	Student	Form is simple; status of registrations is easy to see.	Added coloured status labels in the report
Test user 2	Event Coordinator	Only assigned registrations are shown, which is clear.	None required
Test user 3	Administrator	Dashboard gives a quick summary of open registrations.	Added quick action links

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed in the Development environment, promoted to Stage for acceptance testing and finally published to Production. Changes are always made in Development first, so the live application used by students is not disturbed. Figure 7.1 shows the three environments with their versions, and Table 7.1 lists the checks completed before publishing.

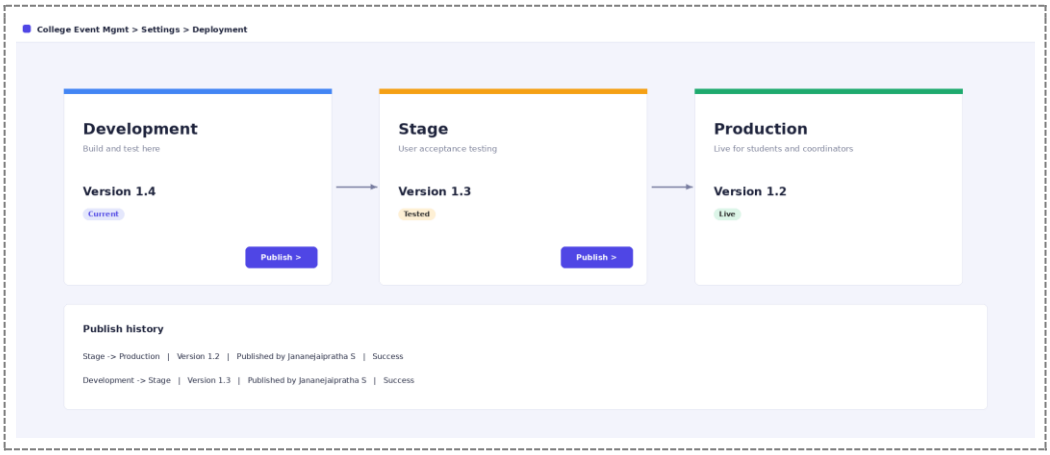


Figure 7.1: Environments and Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Role permissions tested with separate accounts	✓
4	Email delivery tested with authorised addresses	✓
5	Connections authorised with minimum scopes	✓
6	Reports and dashboard reconciled with records	✓
7	Backup / export procedure reviewed	✓
8	Application published to Production	✓

7.2 Performance Optimisation

Performance was kept acceptable by design choices rather than by heavy tuning. Scripts fetch only the single student or category record that they need, reports use filters and criteria to open focused views, and the schedule processes only records that are overdue. Table 7.2 lists the measures applied. Timing figures are not claimed because the sample dataset is small.

Table 7.2: Performance Optimisation Measures

Optimisation Applied	Component	Approach	Expected Benefit
Criteria-based fetch	DS-01 to DS-04	Fetch by record ID instead of looping over all records	Fewer records read per submission
Report filters and criteria	Open Registrations, Pending Payments	Show only records needed by the role	Faster loading of focused views
Criteria in schedule	DS-06	Select only overdue, unattended registrations	Shorter nightly processing
Dashboard from reports	Event Management Dashboard	Charts use existing report criteria	No duplicate calculations

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based permissions	Administrator, Manager, Coordinator and Student profiles	Unauthorised access to records and actions
Record-level criteria	Student and Coordinator views of registrations	One user reading another user's data
Mandatory fields and validation	Student, registration, payment and feedback forms	Incomplete or invalid data
OAuth connections with minimum scopes	cliq_alert_conn, creator_api_conn	Exposure of passwords or excessive API access
Recipient checks	Email workflows	Sending messages to empty or invalid recipients
Hosted over HTTPS	Zoho Creator service	Interception of data in transit
Audit trail	Administration settings	Untraceable configuration or data changes

Security controls follow the least-privilege principle: each role is given only the access that its tasks require. Student contact information is shown only to authorised

coordinator, and email messages contain only the information needed for the student to identify the registration.

7.4 Governance and Administration

Governance practices include meaningful form and field names, restricted access to student contact information, review of workflow changes before publishing, periodic export of data as a backup and regular review of the audit trail. Figure 7.2 shows the audit trail listing who added or changed records and configuration.

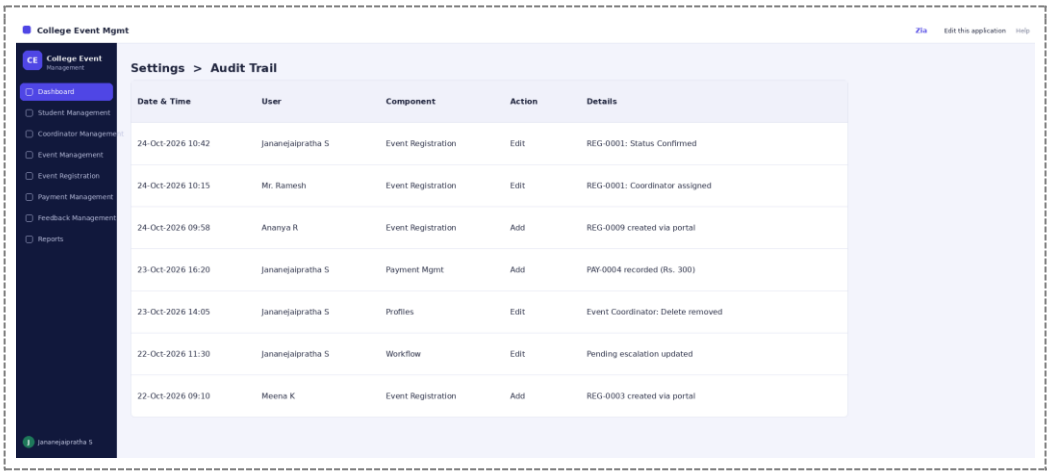


Figure 7.2: Audit Trail and Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A sample end-to-end scenario begins when a student signs in to the portal and submits a speaker technical symposium registration as a speaker (Figure 8.1). The system sets the confirmation deadline from the category confirmation time, sends an acknowledgement email and posts a Zoho Cliq alert. The administrator then assigns the registration to the event coordinator, who receives an assignment email (Figure 8.2). When the event coordinator marks the registration as Attended, the student receives a attendance email with a feedback link (Figure 8.3), and the feedback comment is analysed by Zia.

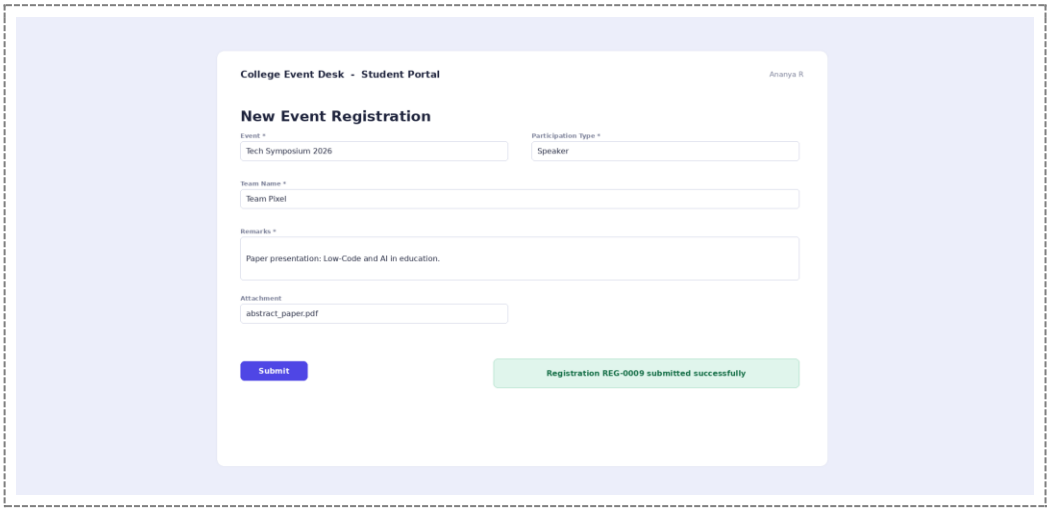


Figure 8.1: Scenario Step 1 – Student Submits an Event Registration in the Portal

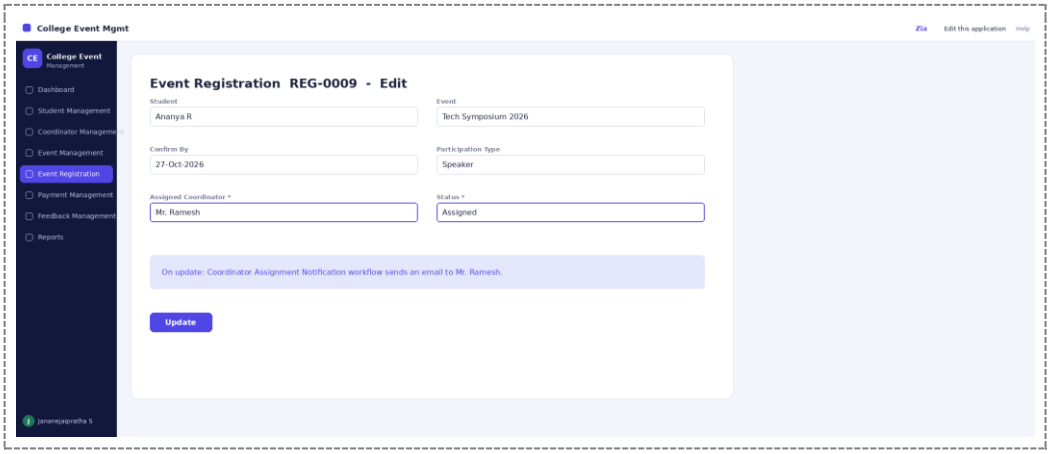


Figure 8.2: Scenario Step 2 – Administrator Assigns the Registration to Coordinator

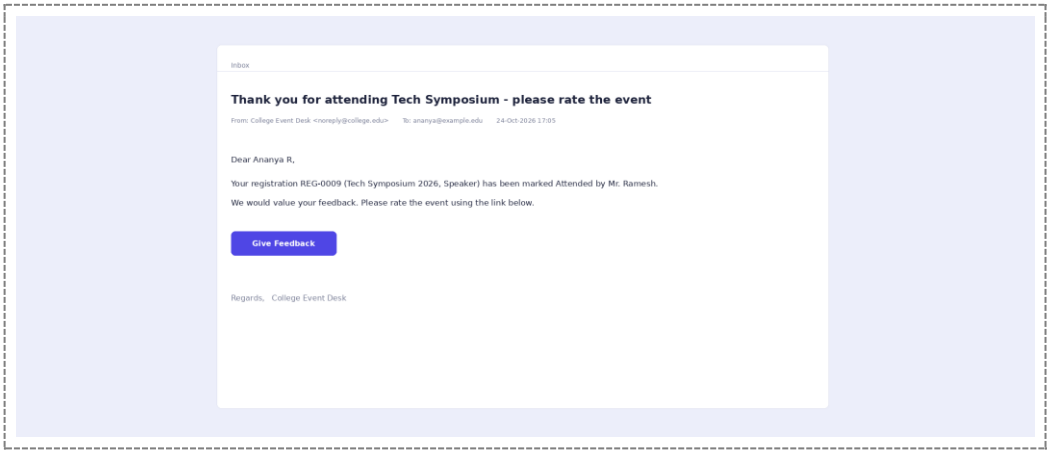


Figure 8.3: Scenario Step 3 – Attendance Email Received by the Student

The scenario demonstrates the full registration lifecycle: register, acknowledge, assign, confirm, attend and collect feedback, with the dashboard and reports updating automatically at each step.

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Create related records for campus events	Sections 4.2 and 5.2	Six forms with lookup relationships (Figs. 4.2, 5.2–5.4).
Automate validation, confirmation deadlines and notifications	Sections 5.5 and 5.6	Scripts DS-01 to DS-06 and eight workflows tested in Table 6.1.
Provide reports and dashboard	Sections 5.3–5.4	Six reports, summary tiles and four charts (Figs. 5.5–5.7).
Integrate with Zoho Cliq / REST API and use Zia	Sections 5.7 and 5.9	Two OAuth connections, Cliq alert, Postman response and Zia fields (Figs. 5.9, 5.10, 5.13).
Apply role-based access	Sections 5.8 and 7.3	Four roles, profiles and record-level criteria (Figs. 5.11, 5.12).
Deploy securely	Sections 7.1 and 7.4	Development, Stage and Production environments; audit trail (Figs. 7.1, 7.2).

The application consolidates the main records needed for campus event handling and provides linked forms, reports, a dashboard, validation logic, automated notifications, integrations, AI-assisted feedback analysis and role-based access.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Registration registration	Paper slip or register entry at the office	Online form through portal, web or mobile	Registrations can be raised any time and are never lost
Status visibility	Student must visit or call	Student sees status in My Registrations	Transparent tracking
Deadlines	Not defined or tracked	Confirmation deadline from confirmation time; daily overdue escalation	Delays are detected automatically
Notifications	Manual calls or messages	Workflow emails and Cliq alerts	Faster communication
Reporting	Manual counting from registers	Reports, dashboard and charts	Instant summaries for management
Feedback	Rarely collected	Rating, comment and Zia sentiment	Event quality can be measured

Compared with a register-and-spreadsheet process, the proposed application centralises related records and provides reusable reports and automation. The improvement is qualitative; no time-saving percentage is claimed because a before-and-after timing study was not performed.

8.4 Limitations of the Low-Code Approach

The application depends on Zoho Creator's supported features and the account edition. Complex integrations, advanced payment verification, extensive custom user interfaces and very large data volumes may need pro-code extensions. Platform limits on API calls and storage must also be considered, and the application cannot be moved easily to another platform.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The project illustrates the no-code, low-code and pro-code spectrum. Visual form, report and dashboard builders provide much of the application structure without code, Deluge scripts and connections add custom logic and integration, and pro-code tools such as Catalyst remain available for needs beyond the platform. Choosing the lowest level that satisfies a requirement kept the project fast to build and simple to maintain.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The College Event Management System brings student, coordinator, event, registration, payment and feedback data into one Zoho Creator application. It uses lookup relationships, validation, Deluge scripts, workflows, an approval process and a Blueprint to manage the full life of an event registration. Reports, a dashboard and charts give administrators clear visibility, while connections to Zoho Cliq and the Creator REST API, Zia AI fields and role-based access extend the application beyond simple data entry. The project shows that a practical, secure and automated campus event desk can be built quickly with a low-code platform.

9.2 Future Enhancements

- Enhancement 1 – Online payment-gateway integration with automatic receipt generation for paid events.
- Enhancement 2 – Push notifications and WhatsApp or SMS alerts for status changes.
- Enhancement 3 – Integration with the college ERP for automatic student data import.
- Enhancement 4 – Predictive analytics for expected registration volumes and coordinator planning using Zoho Analytics.

9.3 Learning Outcomes and Career Relevance

The project develops practical skills in relational data modelling through lookup fields, low-code form and report design, Deluge scripting, workflow automation, secure API integration, role-based access, AI-assisted features, testing and deployment. These skills are directly relevant to roles in business application development, low-code engineering and enterprise automation.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	College Event Management System	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Student, Event Registration, Payment and Feedback forms	5.2	Figs. 5.2–5.4
II	Lookup relationships between forms	Y	Student, category, coordinator and registration lookups	4.2, 5.2	Fig. 4.2
II	Subform where applicable	N	Not required; payments and feedback are separate linked forms	4.2	Justified in 4.2
II	Reports with filtering and sorting	Y	Six reports (List, Kanban, Summary)	5.3	Fig. 5.5
II	Dashboard / page with summaries	Y	Event Management Dashboard	5.4	Figs. 5.6–5.7
II	Deluge validation / calculation scripts	Y	DS-01 validation and due-date calculation	5.5	Code 5.1
II	At least three meaningful workflows	Y	Five form workflows, schedule, approval, Blueprint	5.6	Fig. 5.8
II	Email notifications tested end-to-end	Y	Acknowledgement, assignment, attendance emails	6.2, 8.1	Fig. 8.3
III	Roles and permissions	Y	Four roles with profiles	5.8, 7.3	Figs. 5.11–5.12
IV	External integration / API	Y	Zoho Cliq alert via connection	5.7	Fig. 5.9, Code 5.4
V	Security and governance	Y	Permissions, validation, audit trail	7.3–7.4	Fig. 7.2
II	Form workflow	Y	DS-01 to DS-05	5.6	Fig. 5.8
II	Scheduled workflow	Y	Pending Registration Escalation (daily 9:00 AM)	5.6	Code 5.3
II	Approval process	Y	Inter-College Event Approval	5.6	Table 5.3

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Blueprint	Y	Registration Lifecycle	5.6	Table 5.3
II	Batch / report workflow	N	Not required for this scope	5.6	Justified in 5.6
II	Testing & debugging evidence	Y	Test cases and debug output	6	Table 6.1, Fig. 6.1
III	Integration with a Zoho service	Y	Zoho Cliq	5.7	Fig. 5.9
III	Connection / REST API call (invokeurl)	Y	DS-05 using cliq_alert_conn	5.7	Code 5.4
III	Webhook or Creator REST API	Y	Creator REST API v2 tested in Postman	5.7	Fig. 5.10
III	Users, roles and permission profiles	Y	Roles, users and profiles	5.8	Figs. 5.11–5.12
III	Record-level access criteria / customer portal	Y	Student portal with own-record criteria	5.8, 5.11	Fig. 5.14
III	Zia / AI feature	Y	Sentiment and keyword AI fields	5.9	Fig. 5.13
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts (Zoho Analytics not used)	5.4, 5.9	Fig. 5.7
IV	Performance optimisation applied & measured	Y	Measures applied; timings not claimed	7.2	Table 7.2
IV	Deployment across environments	Y	Development, Stage, Production	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	OAuth connections, HTTPS, profiles	7.3	Table 7.3
IV	Governance (audit trail, administration)	Y	Audit trail review	7.4	Fig. 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not required; justified in 5.10	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	—
—	Mobile app / portal access	Y	Student portal and mobile app	5.11	Fig. 5.14

Total features implemented: 29 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application platforms.	Chapters 1–2	3	
CO2	Develop business applications using forms, reports, workflows and Deluge.	Chapters 4–6	3	
CO3	Apply integration, access control, security and deployment concepts.	Chapters 5 and 7	3	
CO4	Test, debug and document a low-code application.	Chapters 6 and Appendices	3	
CO5	Evaluate the application and identify practical future enhancements.	Chapters 8–9	3	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Event Registration – Validation and Due-Date Script (DS-01)

```
// Script ID : DS-01
// Location  : Event Registration > Workflow > On Add > On Validate
// Purpose   : Validate registration date and confirm days, then set
the confirmation deadline

cat = Event[ID == input.Event];
if(input.Registration_Date == null)
{
    alert "Registration Date is required";
    cancel submit;
}
if(cat.Confirm_Days == null || cat.Confirm_Days < 1)
{
    alert "confirm days are not defined for this event";
    cancel submit;
}
input.Confirm_By =
input.Registration_Date.addDay(cat.Confirm_Days.toLong());
```

C.2 Event Registration – Acknowledgement Email Script (DS-02)

```
// Script ID : DS-02
// Location  : Event Registration > Workflow > On Add > On Success
// Purpose   : Email the student that the registration is registered

stu = Student_Management[ID == input.Student];
cat = Event[ID == input.Event];
if(stu.Email != "")
{
    sendmail
    [
        from :zoho.adminuserid
        to :stu.Email
        subject :"Registration " + input.Registration_ID + " received"
        message :"Dear " + stu.Student_Name + ", your " +
cat.Event_Name + " registration is registered. Confirmation deadline:
" + input.Confirm_By
    ]
}
```

C.3 Event Registration – Coordinator Assignment Notification (DS-03)

```
// Script ID : DS-03
// Location  : Event Registration > Workflow > On Edit > On Success
// Purpose   : Notify the assigned coordinator

if(input.Assigned_Coordinator != null && input.Status == "Assigned")
{
    coordinator = Coordinator_Management[ID ==
input.Assigned_Coordinator];
    sendmail
    [
        from :zoho.adminuserid
        to :coordinator.Email
        subject :"New registration assigned: " + input.Registration_ID
        message :"Registration " + input.Registration_ID + " (" +
input.Team Name + ") is assigned to you. Confirmation deadline: " +
input.Confirm_By
    ]
}
```

C.4 Event Registration – Attendance and Feedback Invitation (DS-04)

```
// Script ID : DS-04
// Location  : Event Registration > Workflow > On Edit > On Success
// Purpose   : Tell the student the registration is attended and
invite feedback

if(input.Status == "Attended")
{
    stu = Student_Management[ID == input.Student];
    if(stu.Email != "")
    {
        sendmail
        [
            from :zoho.adminuserid
            to :stu.Email
            subject :"Registration " + input.Registration_ID + "
attended - please rate the event"
            message :"Dear " + stu.Student_Name + ", your registration
has been attended. Please give feedback in the student portal."
        ]
    }
}
```

C.5 Event Registration – High Participation Type Zoho Cliq Alert (DS-05)

```
// Script ID : DS-05
// Location  : Event Registration > Workflow > On Add > On Success
// Purpose   : Post an alert in Zoho Cliq using the cliq_alert_conn
connection

if(input.Participation_Type == "Speaker")
{
    payload = Map();
    payload.put("text", "Speaker registration " +
input.Registration_ID + ": " + input.Team Name);
    response = invokeurl
    [
        url
:"https://cliq.zoho.in/api/v2/channelsbyname/collegeevents/message"
        type :POST
        parameters :payload.toString()
        connection :"cliq_alert_conn"
    ];
    info response;
}
```

C.6 Schedule – Pending Registration Escalation (DS-06)

```
// Script ID : DS-06
// Location  : Schedule > Pending Registration Escalation (daily 9:00
AM)
// Purpose   : Email the administrator about overdue, unattended
registrations

for each rec in Event_Registration[Confirm_By < zoho.currentdate &&
Status != "Attended" && Status != "Closed"]
{
    sendmail
    [
        from :zoho.adminuserid
        to :zoho.adminuserid
        subject :"Overdue registration " + rec.Registration_ID
        message :"Registration " + rec.Registration_ID + " (" +
rec.Team Name + ") is past its confirmation deadline " +
rec.Confirm_By
    ]
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open the Zoho Creator application link provided by the administrator and sign in with your own account.

2. Use the left navigation menu to open the forms and reports allowed for your role.

D.2 Instructions for Students

1. Sign in to the student portal and open Event Registration.

2. Select the event, enter a subject and description, choose the participation type and attach a file if needed.

3. Click Submit. A confirmation email is sent and the registration appears in My Registrations with its status and confirmation deadline.

4. When the registration is attended, open the link in the attendance email and give a rating and comment in Feedback.

D.3 Instructions for Event Coordinator

1. Sign in and open the All Event Registrations report; only registrations assigned to you are shown.

2. Open a registration, change the status to In Progress and add notes when work starts.

3. When the work is complete, change the status to Attended so that the student is notified.

4. For paid events, open Payment Management and record the payment against the registration.

D.4 Instructions for the Administrator

1. Maintain students, coordinator and events (including confirm days and fees) in their forms.

2. Open the Open Registrations report, assign coordinator and set the participation type; inter-college event registrations wait for manager approval.

3. Review the Event Management Dashboard daily for open, overdue and speaker registrations.

4. Manage users, roles and profiles from Users & Control, and review the audit trail when needed.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	JANANEJAIPRATHA S	Register No.	241501072
Project Title	College Event Management System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	JANANEJAIPRATHA S	Register No.	241501072
Department / Section	AIML	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____